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ORIGINAL ARTICLE

Influence of *DPYD* gene polymorphisms on 5-Fluorouracil toxicities in Thai colorectal cancer patients

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Abstract

DPYD polymorphisms have been widely found to be related to 5-FU-induced toxicities. The aim of this study was to establish significant associations between five single-nucleotide polymorphisms of *DPYD* and 5-FU hematological toxicities in Thai colorectal cancer patients. The toxicities were analyzed at the first and second cycles of 5-FU administration in 75 patients. Genotyping was performed using TaqMan real-time PCR. The genotype frequencies of *DPYD**2A,1905 + 1 G > A and *DPYD* 1774 C > T were all wild type. The frequencies of genetic testing for *DPYD**5, 1627 A > G, *DPYD* 1896T > C, and *DPYD**9A, 85 A > G were 37.30% (AG; 34.60%, GG; 2.70%), 32.00% (TC; 25.30%, CC; 6.70%), and 13.40% (AG; 10.70%, GG; 2.70%), respectively. The results reveal significant findings with neutropenia occurring in 100% (2/2) of the patients with homozygous variant *DPYD**9A (GG) from the first cycle of treatment for both Grade 1–4 and Grade 3–4 toxicities ($P = 0.003$ and $P < 0.001$ respectively). *DPYD* *9A was related to Grade 1–4 leukopenia ($P = 0.001$) and both Grade 1–4 and severe thrombocytopenia ($P < 0.001$ and $P < 0.001$) in the first cycle. In the second cycle, *DPYD**5 was shown to be closely associated with no Grade 1–4 toxicity ($P = 0.02$). However, we found that 100% (2/2) of patients carrying the homozygous variant (GG) *DPYD**5, presented no significant toxicity, so, *DPYD**5 may be a predictive marker of neutropenia in patients treated with 5-FU. These outcomes suggest that there may be an increased risk of developing 5-FU-induced neutropenia in patients carrying the *DPYD**9A, which should be considered as part of the standard procedure.

Keywords *DPYD* polymorphisms · Toxicity · 5-Fluorouracil · Colorectal cancer

Introduction

The anticancer drug, 5-Fluorouracil (5-FU) a common choice for treatment of various cancers, including colorectal cancer. Its common side effects include severe neutropenia and severe diarrhea, which can lead the body to enter a shock state; however, side effects of 5-FU are life-threatening in a minor proportion of patients [1, 2].

This adverse reaction has been found to be closely associated with a decreased function of the *DPYD* gene, which is responsible for the production of the dihydropyrimidine dehydrogenase (DPD) enzyme that metabolizes fluoropyrimidines, the class of antimetabolites to which 5-FU belong [3]. Therefore, this study focuses on the *DPYD* pathway which is responsible for 80–90% of the elimination of 5-FU [4].

The correlation between *DPYD* single-nucleotide polymorphisms (SNPs) and the life-threatening adverse side effects produced when receiving 5-FU-based chemotherapy have been suggested in relevant by existing studies. One study found that grade 3–4 diarrhea could be induced by *DPYD* *9A, 85 A > G (rs1801265) [5] while another study stated that bone marrow depression and neurotoxicity were found to be associated with *DPYD* *2A, 1905 + 1 G > A (rs3918290), *DPYD* *13, 1679 T > G (rs55886062) and *DPYD* 2846 A > T (rs67376798) [6]. *DPYD* variant patients

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have been shown to present increased treatment-related death when prescribed 5-FU chemotherapy [7]. Despite the existing evidence, testing for *DPYD* SNPs has yet to be officially recommended or required by the FDA prior to initiating 5-FU treatment in cancer patients [8].

The aim of this study is to explore any association existing between the genetic polymorphisms of the *DPYD* gene, including *DPYD* *5, 1627 A > G (rs1801159), *DPYD* *9A, 85 A > G, *DPYD* 1896 T > C (rs17376848), *DPYD* *2A, 1905 + 1 G > A, *DPYD* 1774 C > T (rs59086055), and the varying hematological toxicity as a side effect of patients receiving 5-FU chemotherapy for colorectal cancer, in order to better prescribe 5-FU in patients with known *DPYD* variants.

Methods

Sample collection

A total of 75 Thai colorectal cancer patients were recruited between October 2020 and October 2021 from the Division of Oncology, Department of Medicine, Faculty of Medicine, Ramathibodi Hospital, Mahidol University, Thailand. The clinical qualifications used to enlist patients comprised the following inclusion and exclusion criteria: the inclusion criteria included a histologically confirmed invasive colorectal cancer, being over 18 years of age, an ECOG status of 0–2, no previous treatment with 5-FU, a total WBC count of $\geq 3.5 \times 10^9$/L, an absolute neutrophil count of $\geq 1.5 \times 10^9$/L, a hemoglobin level of ≥ 9 g/dL, a platelet count of $\geq 8 \times 10^{10}$/L, an AST & ALT level of $\geq 8 \times 10^{10}$/L, and a creatinine level of ≤ 125 mg/dL. The exclusion criteria included pregnancy and any laboratory evidence of renal or hepatic abnormalities. All patients received 5-FU-based chemotherapy for adjuvant treatment; their drug regimens are detailed in Table 1.

Alongside the criteria above, single-nucleotide polymorphism (SNP) genotyping was conducted for drug-metabolizing enzymes using TaqMan real-time PCR. Genotyping was performed for five known SNPs of the *DPYD* gene, *DPYD* *5,1627 A > G (rs1801159), *DPYD* *9A 85 A > G (rs1801265), *DPYD* 1896 T > C (rs17376848), *DPYD* *2A,1905 + 1 G > A (rs3918290), and *DPYD* 1774 C > T (rs59086055).

This study was approved by the Ethics Review Committee on Human Research of the Faculty of Medicine Ramathibodi Hospital, Mahidol University, Thailand

Table 1 Drug administration

Regimen	Dose	No. of patients (%)
FOLFOX	Intravenous oxaliplatin 85 mg/m^2 on day 1 and a 2-hour infusion of leucovorin 200 mg/m^2 followed by bolus 5-FU 400 mg/m^2 and a 22-hour continuous infusion of 5-FU 600 mg/m^2 for two consecutive days with treatment repeated every two weeks	21 (28.00)
5-Fluorouracil + Leucovorin	5-FU at 425 mg/m^2/day, 5 days + leucovolin 30 mg	21 (28.00)
FOLFIRI	Irinotecan 180 mg/m2, 90 min intravenous infusion on day 1; leucovorin (LV) 200 mg/m^2 intravenous infusion on day 1; fluorouracil 400 mg/m^2 intravenous bolus on day 1; fluorouracil 600 mg/m^2 intravenous over the course of 46 h of continuous infusion; repeated every 2 weeks	13 (17.30)
Modified FOLFOX	Intravenous oxaliplatin 85 mg/m^2 on day 1 and a 2-hour infusion of leucovorin 200 mg/m^2 followed by bolus 5-FU 400 mg/m^2 and a 22-hour continuous infusion of 5-FU 1200 mg/m^2 for two consecutive days with treatment repeated every two weeks	7 (9.30)
Modified FOLFIRI	Irinotecan 180 mg/m2, 90 min intravenous infusion on day 1; leucovorin (LV) 400 mg/m2 intravenous infusion on day 1; fluorouracil 400 mg/m2 intravenous bolus on day 1; fluorouracil 1,200 mg/m2 intravenous over the course of 46 h of continuous infusion; repeated every 2 weeks	6 (8.00)
FOLFIRI + AVASTIN	Avastin 5–10 mg/kg intravenous infusion once every 2 weeks; Irinotecan 180 mg/m2, 90 min intravenous infusion on day 1; leucovorin (LV) 200 mg/m2 intravenous infusion on day 1; fluorouracil 400 mg/m2 intravenous bolus on day 1; fluorouracil 600 mg/m2 intravenous over the course of 46 h of continuous infusion; repeated every 2 weeks	3 (4.00)
FOLFOX + AVASTIN	Avastin 5–10 mg/kg intravenous infusion once every 2 weeks; intravenous oxaliplatin 85 mg/m^2 on day 1 and a 2-hour infusion of leucovorin 200 mg/m^2 followed by bolus 5-FU 400 mg/m^2 and a 22-hour continuous infusion of 5-FU 600 mg/m^2 for two consecutive days with treatment repeated every two weeks	2 (2.70)
FOLFIRI + ERBITUX	Erbitux 400 mg/m2 intravenous infusion on day 1; Irinotecan 180 mg/m2, 90 min intravenous infusion on day 1; leucovorin (LV) 200 mg/m2 intravenous infusion on day 1; fluorouracil 400 mg/m2 intravenous bolus on day 1; fluorouracil 600 mg/m2 intravenous over the course of 46 h of continuous infusion; repeated every 2 weeks	1 (1.30)
Modified FOLFOX + AVASTIN	Avastin 5–10 mg/kg intravenous infusion once every 2 weeks; intravenous oxaliplatin 85 mg/m^2 on day 1 and a 2-hour infusion of leucovorin 200 mg/m^2 followed by bolus 5-FU 400 mg/m^2 and a 22-hour continuous infusion of 5-FU 1200 mg/m^2 for two consecutive days with treatment repeated every two weeks	1 (1.30)

(MURA2020/1613) and conducted in accordance with the Declaration of Helsinki. The study protocol was clearly explained to all patients and informed consent was given before the study. Informed consent to publication was also provided.

Genotyping analysis

In the recruited 75 patients, 5-FU based chemotherapy was conducted for a total of two cycles, and blood samples were drawn from each patient a total of three times over the course of the study. The first sample was taken before 5-FU administration, serving as a baseline sample while the other two were taken after cycle 1 and cycle 2, respectively. The EDTA blood tube of the baseline sample underwent a pharmacogenetic analysis and a complete blood count (CBC), while the clotted blood tube of the baseline sample was subjected to a clinical chemistry analysis and measurements for levels of AST, ALT, and creatinine. The blood tests performed for blood samples post-cycle 1 and post-cycle 2 were identical; the EDTA blood tube received a CBC and the clotted blood tube completed a clinical chemistry analysis.

Blood samples were collected to perform DNA extraction using MagNA Pure Compact System (Roche, Manheim, Germany) and TaqMan real-time PCR was used to identify 5 SNPs of the *DPYD* gene, namely, *DPYD**5, 1627 A > G (Assay ID: C_1823316_20), *DPYD**9A, 85 A > G (C_9491497_10), *DPYD* 1896T > C (C_25471727_20), *DPYD**2A, 1905 + 1G > A (C_30633851_20), and *DPYD* 1774 C > T (C_90454263_10). This was followed by DNA purification and concentration executed using a Thermo Scientific™ NanoDrop™ spectrophotometer (Thermo Fisher Scientific, DE, USA). The data obtained regarding clinical toxicity were studied for associations with the specified genetic polymorphisms through statistical analysis.

Toxicity criteria

All patients were analyzed for hematological toxicity using the CTCAE v.5.0 assessment criteria. The considered hematological toxicities included the levels of neutropenia, leukopenia, thrombocytopenia and anemia. The toxicity levels were graded with Grade 0 indicating 'No toxicity at all', Grade 1–2 being 'Mild', and Grade 3–4 being 'Severe'. All hematological toxicities were analyzed in the first and second cycles.

Statistical analysis

Descriptive statistics are used to describe the clinical characteristics of the subjects. Data are expressed as mean and standard deviation (SD) for normally distributed data. Conformity to Hardy–Weinberg equilibrium (HWE) was determined using Fisher's exact test. Comparisons of allele and genotype frequencies and grades of toxicity were performed using the $\chi 2$ test. A two-sided P-value less than 0.05 was considered statistically significant. Statistical analyses were performed using the SPSS software package (SPSS version 21.0 for Windows, SPSS Inc, Chicago, IL).

Results

Genotype and allele frequencies of DPYD polymorphisms

As demonstrated in Table 2, all 75 patients were genotyped for *DPYD**5,1627 A > G, *DPYD**9A 85 A > G, *DPYD* 1896 T > C, *DPYD**2A,1905 + 1 G > A, and *DPYD* 1774 C > T. The *DPYD**2A,1905 + 1 G > A and *DPYD* 1774 C > T variants were not found in our patients. The most prevalent variant allele frequency was *DPYD**5,1627 A > G with an allele frequency of 0.20 while *DPYD* 1896 T > C followed closely at 0.19 and *DPYD* *9A 85 A > G was the least prevalent variant, with an allele frequency of 0.08.

Clinical characteristics

The clinical characteristics of the 75 metastatic colorectal cancer patients who received 5-FU are provided in Table 3. The population's average age was 60.30 years with 61.30% (46/75) males and 38.70% (29/75) females. Forty-nine patients (65.30% (49/75)) had an ECOG performance status of zero (which was the majority), while only 2.70% (2/75) had an ECOG performance of two. Most patients 45.30% (34/75) had the disease at the rectum, making it the most prevalent site of disease while the most common site of metastases was the liver (61.33%; 46/75). A majority of

Table 2 Genotype and allele frequencies of 75 metastatic colorectal cancer patients. W wild type, V variant

Gene Polymorphisms	Genotype frequency (%)		
	w/w	w/v	v/v
*DPYD *2A (c.1905 + 1G > A) (rs 3918290)*	75 (100.00%)	0 (0.00%)	0 (0.00%)
*DPYD*5 (1627 A > G) (rs 1801159)*	47 (62.70%)	26 (34.60%)	2 (2.70%)
DPYD c.1774 C > T (rs 59086055)	75 (100.00%)	0 (0.00%)	0 (0.00%)
*DPYD*9A c.85 A > G (rs 1801265)*	65 (86.60%)	8 (10.70%)	2 (2.70%)
DPYD 1896 T > C (rs 17376848)	51 (68.00%)	19 (25.30%)	5 (6.70%)

Table 3 Clinical characteristics of the patients with colorectal cancer treated with 5-fluorouracil ($N=75$)

Characteristics	Number of patients (%)
Age (years), mean ± SD	60.30 ± 1.20
Gender	
Male	46 (61.30)
Female	29 (38.70)
ECOG performance status	
0	49 (65.30)
1	24 (32.00)
2	2 (2.70)
Site of disease	
Rectum	34 (45.30)
Sigmoid	20 (26.70)
Right side	9 (12.00)
Rectosigmoid	5 (6.70)
Transverse	3 (4.00)
Left side	4 (5.30)
Sites of metastases	
Liver	29 (38.67)
Others	22 (29.33)
Liver and lung	17 (22.67)
Lung	7 (9.33)
Histopathology type	
Well differentiated	17 (22.70)
Moderately differentiated	54 (72.00)
Poorly differentiated	4 (5.30)
Line of treatment	
First line	58 (77.30)
Second line	15 (20.00)
Third line	2 (2.70)
Treatment regimen	
5-Fluorouracil + Leucovorin	21 (28.00)
FOLFOX	21 (28.00)
FOLFIRI	13 (17.30)
Modified FOLFOX	7 (9.30)
Modified FOLFIRI	6 (8.00)
FOLFIRI + AVASTIN	3 (4.00)
FOLFOX + AVASTIN	2 (2.70)
FOLFIRI + ERBITUX	1 (1.30)
Modified FOLFOX + AVASTIN	1 (1.30)

the patients (77.30%; 58/75) were receiving 5-FU in this study as their first-line treatment, while 20.00% (15/75) of patients were receiving it as second-line and 2.70% (2/75) patients were receiving it as third-line. In particular, 28.00% (21/75) of patients received FOLFOX or Leucovorin with 5-FU; these two drug regimens were the most common amongst the sample.

Correlation of clinical characteristics with hematological toxicities

Hematological toxicities were assessed using levels of neutropenia, leukopenia, thrombocytopenia and anemia, shown in Table 4. There were no statistically significant differences between clinical characteristics, and leukopenia, thrombocytopenia or anemia. However, some clinical characteristics were significantly associated with neutropenia toxicity. An age of < 60 years old, female sex, ECOG status 2, lung metastases, and 5-FU as a third-line treatment were associated with neutropenia toxicity, as shown in Table 5.

Correlation of *DPYD* gene polymorphisms with hematological toxicities

DPYD polymorphisms were determined with hematological toxicities including anemia, leucopenia, and thrombocytopenia as showed in Table 6. *DPYD**9A, 85 A > G was related to Grade 1–4 leucopenia at first and second cycles ($P=0.001$, and $P=0.007$ respectively), and both Grade 1–4 and Grade 3–4 thrombocytopenia at first cycle ($P<0.001$, and $P<0.001$respectively). As illustrated in Table 7, some significance was demonstrated with regard to neutropenia toxicity. The results show that the *DPYD**5, 1627 A > G variant presented a statistically significant difference at the second cycle of 5-FU ($P=0.02$), when comparing the grade 1–4 toxicity groups against the grade 0 group. Furthermore, *DPYD**9A, 85 A > G was found to be statistically significant from the first cycle of 5-FU for both toxic (Grade 1–4) and non-toxic group comparisons, as well as the severe toxicity group (Grade 3–4) ($P=0.003$, and $P<0.001$ respectively). It should be noted that patients carrying the homozygous variant (G/G) of *DPYD**9A, 85 A > G had a rate of development of Grade 3–4 neutropenia toxicity from the first cycle of 5-FU. Table S2 shows the associations between the *DPYD* polymorphisms and leukopenia, thrombocytopenia and anemia. The results indicate that *DPYD**9A, 85 A > G was significantly related to grade 1–4 leukopenia in both the first

Table 4 Hematological toxicities including leukopenia, neutropenia, thrombocytopenia, and anemia in first and second cycles ($N=75$)

Toxicity	N	First cycle					Second cycle				
		Grade 0 n (%)	Grade 1 n (%)	Grade 2 n (%)	Grade 3 n (%)	Grade 4 n (%)	Grade 0 n (%)	Grade 1 n (%)	Grade 2 n (%)	Grade 3 n (%)	Grade 4 n (%)
Anemia	75	26 (34.70)	35 (46.70)	12 (16.00)	2 (2.70)	0 (0.00)	29 (38.70)	36 (48.00)	9 (12.00)	1 (1.30)	0 (0.00)
Leucopenia	75	53 (70.70)	15 (20.00)	7 (9.30)	0 (0.00)	0 (0.00)	43 (57.30)	27 (36.00)	4 (5.30)	1 (1.30)	0 (0.00)
Neutropenia	75	50 (66.70)	12 (16.00)	2 (2.70)	8 (10.70)	3 (4.00)	65 (86.70)	8 (10.70)	1 (1.30)	1 (1.30)	0 (0.00)
Thrombocytopenia	75	72 (96.00)	2 (2.70)	0 (0.00)	1 (1.30)	0 (0.00)	67 (89.30)	6 (8.00)	0 (0.00)	1 (1.30)	1 (1.30)

Table 5 Correlation of clinical characteristics with neutropenia in first and second cycles ($N=75$)

Clinical characteristics	Toxicity (neutropenia)											
	First cycle						Second cycle					
	Grade 0	Grade 1-4[a]	**P**	Grade 0–2	Grade 3-4[b]	**P**	Grade 0	Grade 1-4[a]	**P**	Grade 0–2	Grade 3-4[b]	**P**
Age (years)												
≥ 60 ($n=43$)	29 (67.40%)	14 (32.60%)	0.840	36 (83.70%)	7 (16.30%)	0.545	22 (51.20%)	21 (48.80%)	0.845	39 (90.70%)	4 (9.30%)	0.017*
<60 ($n=32$)	21 (65.60%)	11 (34.40%)		28 (87.50%)	4 (12.50%)		17 (53.10%)	15 (46.90%)		24 (75.00%)	8 (25.00%)	
Gender												
Male ($n=46$)	34 (73.90%)	12 (26.10%)	0.043*	43 (93.50%)	3 (6.50%)	0.001*	25 (54.30%)	21 (45.70%)	0.553	40 (87.00%)	6 (13.00%)	0.251
Female ($n=29$)	16 (55.20%)	13 (44.80%)		21 (72.4%)	8 (27.60%)		14 (48.30%)	15 (51.70%)		23 (79.30%)	6 (20.70%)	
ECOG performance status												
0 ($n=49$)	32 (65.30%)	17 (34.70%)	0.710	41 (83.70%)	8 (16.30%)	0.597	27 (55.10%)	22 (44.90%)	0.691	39 (79.60%)	10 (20.40%)	0.016*
1 ($n=24$)	6 (70.8%)	7 (29.20%)		21 (87.50%)	3 (12.50%)		11 (45.80%)	13 (54.20%)		23 (95.80%)	1 (4.20%)	
2 ($n=2$)	1 (50.00%)	1 (50.00%)		2 (100.00%)	0 (0.00%)		1 (50.00%)	1 (50.00%)		1 (50.00%)	1 (50.00%)	
Site of disease												
Rectum ($n=34$)	22 (64.70%)	12 (35.30%)	0.963	30 (88.20%)	4 (11.80%)	0.528	15 (44.10%)	19 (55.90%)	0.404	27 (79.40%)	7 (20.60%)	0.238
Sigmoid ($n=20$)	13 (65.00%)	7 (35.00%)		16 (80.00%)	4 (20.00%)		12 (60.00%)	8 (40.00%)		18 (90.00%)	2 (10.00%)	
Right side ($n=9$)	6 (66.70%)	3 (33.30%)		8 (88.90%)	1 (11.10%)		4 (44.40%)	5 (55.60%)		9 (100.00%)	0 (0.00%)	
Rectosigmoid ($n=5$)	4 (80.00%)	1 (20.00%)		4 (80.00%)	1 (20.00%)		4 (80.00%)	1 (20.00%)		4 (80.00%)	1 (20.00%)	
Transverse ($n=3$)	2 (66.70%)	1 (33.30%)		2 (66.70%)	1 (33.30%)		2 (66.70%)	1 (33.30%)		2 (66.70%)	1 (33.30%)	
Left side ($n=4$)	0 (0.00%)	4 (100.00%)		4 (100.00%)	0 (0.00%)		2 (50.00%)	2 (50.00%)		3 (75.00%)	1 (25.00%)	
Sites of metastases												
Liver ($n=29$)	19 (65.50%)	10 (34.50%)	0.002*	26 (89.70%)	3 (10.30%)	0.002*	14 (48.30%)	15 (51.70%)	0.011*	26 (89.70%)	3 (10.30%)	0.437
Others ($n=22$)	12 (54.50%)	10 (45.50%)		17 (77.30%)	5 (22.70%)		11 (50.00%)	11 (50.00%)		18 (81.80%)	4 (18.20%)	
Liver and lung ($n=17$)	16 (94.10%)	1 (5.90%)		17 (100.00%)	0 (0.00%)		13 (76.50%)	4 (23.50%)		14 (82.4%)	3 (17.60%)	
Lung ($n=7$)	3 (42.90%)	4 (57.10%)		4 (57.10%)	3 (42.90%)		1 (14.30%)	6 (85.70%)		5 (71.40%)	2 (28.60%)	
Histopathology type												
Well differentiated ($n=17$)	10 (58.80%)	7 (41.20%)	0.397	13 (85.30%)	4 (14.70%)	0.196	11 (52.00%)	6 (48.00%)	0.215	16 (94.10%)	1 (5.90%)	0.196
Moderately differentiated ($n=54$)	38 (70.40%)	16 (29.60%)		48 (88.90%)	6 (11.10%)		27 (50.00%)	27 (50.00%)		44 (81.50%)	10 (18.50%)	
Poorly differentiated ($n=4$)	2 (50.00%)	2 (50.00%)		3 (75.00%)	1 (25.00%)		1 (25.00%)	3 (75.00%)		3 (75.00%)	1 (25.00%)	
Line of treatment												
First line ($n=58$)	35 (60.30%)	23 (39.70%)	0.004*	48 (82.80%)	10 (17.20%)	0.266	29 (50.00%)	29 (50.00%)	0.713	47 (81.00%)	11 (19.00%)	0.200
Second line ($n=15$)	14 (93.30%)	1 (6.70%)		14 (93.30%)	1 (6.70%)		9 (60.00%)	6 (40.00%)		14 (93.30%)	1 (6.70%)	
Third line ($n=2$)	1 (50.00%)	1 (50.00%)		2 (100.00%)	0 (0.00%)		1 (50.00%)	1 (50.00%)		2 (100.00%)	0 (0.00%)	
Regimen												
Include 5-FU ($n=52$)	36 (69.23%)	16 (30.77%)	0.385	44 (84.62%)	8 (15.38%)	0.741	30 (57.69%)	22 (42.31%)	0.067	45 (86.54%)	7 (13.46%)	0.280
Include 5-FU + irinotecan ($n=23$)	14 (60.87%)	9 (39.13%)		20 (86.96%)	3 (13.04%)		9 (39.13%)	14 (60.87%)		18 (78.26%)	5 (21.74%)	

ND not determined. **P* value was considered statistically significant. [a] Grade 1–4 was considered as toxicity. [b] Grade 3–4 was considered as severe toxicity. Data analyzed with the chi-square test

Table 6 Correlation of *DPYD* gene polymorphisms with leukopenia, thrombocytopenia and anemia in first and second cycles ($N=75$)

Toxicity	Gene	Genotype	N	First cycle				Second cycle			
				Grade 1-4[a] n (%)	P	Grade 3-4[b] n (%)	P	Grade 1-4[a] n (%)	P	Grade 3-4[b] n (%)	P
Anemia	*DPYD *2A (c.1905 + 1G > A)*	GG	75	49 (65.33)	ND	2 (2.67)	ND	46 (61.33)	ND	1 (1.33)	ND
	*DPYD*5 (c.1627 A > G)*	A/A	47	31 (65.96)	0.171	1 (2.13)	0.799	30 (63.83)	0.078	0 (0.00)	0.354
		A/G	26	16 (61.54)		1 (3.85)		14 (53.85)		1 (3.85)	
		G/G	2	2 (100.00)		0 (0.00)		2 (100.00)		0 (0.00)	
	DPYD 1774 C > T	C/C	75	49 (65.33)	ND	2 (2.67)	ND	46 (61.33)	ND	1 (1.33)	ND
	*DPYD*9A (c.85 A > G)*	A/A	65	42 (64.62)	0.197	1 (1.54)	0.088	40 (61.54)	0.849	1 (1.54)	0.843
		A/G	8	5 (62.50)		1 (12.50)		5 (62.50)		0 (0.00)	
		G/G	2	2 (100.00)		0 (0.00)		1 (50.00)		0 (0.00)	
	DPYD 1896T > C	T/T	51	33 (64.71)	0.837	1 (1.96)	0.505	32 (62.75)	0.897	1 (1.96)	0.592
		T/C	19	13 (68.42)		1 (5.26)		11 (57.89)		0 (0.00)	
		C/C	5	3 (60.00)		0 (0.00)		3 (60.00)		0 (0.00)	
Leucopenia	*DPYD *2A (c.1905 + 1G > A)*	GG	75	22 (29.33)	ND	0 (0.00)	ND	32 (42.67)	ND	1 (1.33)	ND
	*DPYD*5 (c.1627 A > G)*	A/A	47	16 (34.04)	0.147	0 (0.00)	ND	23 (48.94)	0.045*	1 (2.13)	0.536
		A/G	26	6 (23.08)		0 (0.00)		9 (34.62)		0 (0.00)	
		G/G	2	0 (0.00)		0 (0.00)		0 (0.00)		0 (0.00)	
	DPYD 1774 C > T	C/C	75	22 (29.33)	ND	0 (0.00)	ND	32 (42.67)	ND	1 (1.33)	ND
	*DPYD*9A (c.85 A > G)*	A/A	65	18 (27.69)	0.001*	0 (0.00)	ND	28 (43.08)	0.007*	1 (1.54)	0.843
		A/G	8	2 (25.00)		0 (0.00)		2 (25.00)		0 (0.00)	
		G/G	2	2 (100.00)		0 (0.00)		2 (100.00)		0 (0.00)	
	DPYD 1896T > C	T/T	51	13 (25.49)	0.002*	0 (0.00)	ND	22 (43.14)	0.976	1 (1.96)	0.592
		T/C	19	9 (47.37)		0 (0.00)		8 (42.11)		0 (0.00)	
		C/C	5	0 (0.00)		0 (0.00)		2 (40.00)		0 (0.00)	
Thrombocytopenia	*DPYD *2A (c.1905 + 1G > A)*	GG	75	3 (4.00)	ND	1 (1.33)	ND	8 (10.67)	ND	2 (2.67)	ND
	*DPYD*5 (c.1627 A > G)*	A/A	47	1 (2.13)	0.367	1 (2.13)	0.536	5 (10.64)	0.682	0 (0.00)	0.120
		A/G	26	2 (7.69)		0 (0.00)		3 (11.54)		2 (7.69)	
		G/G	2	0 (0.00)		0 (0.00)		0 (0.00)		0	
	DPYD 1774 C > T	C/C	75	3 (4.00)	ND	1 (1.33)	ND	8 (10.67)	ND	2 (2.67)	ND
	*DPYD*9A (c.85 A > G)*	A/A	65	1 (1.54)	< 0.001*	0 (0.00)	< 0.001*	8 (12.31)	0.225	2 (3.08)	0.707
		A/G	8	1 (12.50)		0 (0.00)		0 (0.00)		0 (0.00)	
		G/G	2	1 (50.00)		1 (50.00)		0 (0.00)		0 (0.00)	
	DPYD 1896T > C	T/T	51	3 (5.88)	0.201	1 (1.96)	0.592	5 (9.80)	0.538	1 (1.96)	0.505
		T/C	19	0 (0.00)		0 (0.00)		2 (10.53)		1 (5.26)	
		C/C	5	0 (0.00)		0 (0.00)		1 (20.00)		0 (0.00)	

ND = Not determined; * P value < 0.05 was *considered* statistically *significant*. [a]Grade 1–4 was considered as toxicity. [b]Grade 3–4 was considered as severe toxicity

and second cycles ($P<0.001$ and $P=0.007$, respectively); meanwhile, the *DPYD**9A, 85 A > G variant was also significantly related to grade 1–4 and severe thrombocytopenia in the first cycle ($P<0.001$ and $P<0.001$, respectively). The *DPYD* 1896 $T>C$ was significantly related to grade 1–4 leukopenia in the first cycle ($P=0.002$). Notably, *DPYD**2A,1905 + 1 G > A and *DPYD* 1774 C > T, presented no statistically significant relationships with hematological toxicities.

Table 7 Correlation of *DPYD* gene polymorphisms with neutropenia in first and second cycles ($N=75$)

Polymorphisms	Toxicity (neutropenia)											
	First cycle						Second cycle					
	Grade 0	Grade 1-4[a]	***P***	Grade 0–2	Grade 3-4[b]	***P***	Grade 0	Grade 1-4[a]	**P**	Grade 0–2	Grade 3-4[b]	***P***
*DPYD *2A (c.1905 + 1G > A) (rs 3918290)*												
G/G ($n=75$)	50 (66.70%)	25 (33.30%)	ND	65 (85.3%)	11 (14.70%)	ND	39 (52.00%)	36 (48.0%)	ND	63 (84.00%)	12 (16.00%)	ND
*DPYD*5 (c.1627 A > G) (rs 1801159)*												
A/A ($n=47$)	29 (61.70%)	18 (38.30%)	0.12	38 (80.90%)	9 (19.10%)	0.14	26 (55.30%)	21 (44.70%)	0.02*	40 (85.10%)	7 (14.90%)	0.45
A/G ($n=26$)	19 (73.10%)	7 (26.90%)		24 (92.30%)	2 (7.70%)		11 (42.30%)	15 (57.70%)		21 (80.80%)	5 (19.20%)	
G/G ($n=2$)	2 (100.00%)	0 (0.00%)		2 (100.00%)	0 (0.00%)		2 (100.00%)	0 (0.00%)		2 (100.00%)	0 (0.00%)	
DPYD 1774 C > T (rs 59086055)												
C/C ($n=75$)	50 (66.70%)	25 (33.30%)	ND	64 (85.30%)	11 (14.70%)	ND	39 (52.00%)	36 (48.0%)	ND	63 (84.00%)	12 (16.00%)	ND
*DPYD*9A (c.85 A > G) (rs 1801265)*												
A/A ($n=65$)	45 (69.20%)	20 (30.80%)	0.003*	56 (86.20%)	9 (13.80%)	< 0.001*	33 (50.80%)	32 (49.20%)	0.69	54 (83.10%)	11 (16.90%)	0.51
A/G ($n=8$)	5 (62.50%)	3 (37.50%)		8 (100.00%)	0 (0.00%)		5 (62.50%)	3 (37.50%)		7 (87.50%)	1 (12.50%)	
G/G ($n=2$)	0 (0.00%)	2 (100.00)		0 (0.00%)	2 (100.00%)		1 (50.00%)	1 (50.00%)		2 (100.00%)	0 (0.00%)	
DPYD 1896T > C (rs 17376848)												
T/T ($n=51$)	36 (70.60%)	15 (29.40%)	0.09	46 (90.20%)	5 (9.80%)	0.12	25 (49.00%)	26 (51.00%)	0.62	43 (84.30%)	8 (15.70%)	0.92
T/C ($n=19$)	10 (52.60%)	9 (47.40%)		14 (73.70%)	5 (26.30%)		11 (57.90%)	8 (42.10%)		16 (84.20%)	3 (15.80%)	
C/C ($n=5$)	4 (80.00%)	1 (20.000%)		4 (80.00%)	1 (20.00%)		3 (60.00%)	2 (40.00%)		4 (80.00%)	1 (20.00%)	

ND not determined. **p* value was considered statistically significant. [a] Grade 1–4 was considered as toxicity. [b] Grade 3–4 was considered as severe toxicity. Data analyzed with the chi-square test

A multivariate logistic regression analysis was performed to analyze the influence of age, gender, ECOG performance status, Sites of metastases, Line of treatment, *DPYD*5*, *DPYD*9A* on neutropenia (all grades and severe neutropenia) at first and second cycles. The result showed that there was no significant association of clinical factors and genetic factors with neutropenia (Table 8).

Discussion

This study explored the association between *DPYD* genetic polymorphisms and 5-FU-induced toxicity amongst Thai colorectal cancer patients. The obtained results demonstrated significant associations between neutropenia toxicity and the *DPYD**9A, 85 A>G and *DPYD**5, 1627 A>G polymorphisms.

As the candidate SNPs in this study, we selected *DPYD** 2A,1905+1 G>A, *DPYD**5, 1627 A>G, and *DPYD* 1774 C>T as they had been found to be related with hematological toxicities in previous studies [6, 9–11]. Meanwhile, *DPYD* 1896T>C and *DPYD**9A, 85 A>G were selected from Thai Exome sequencing with allele frequencies higher than 5%.

The *DPYD**9A, 85 A>G SNP is responsible for a missense mutation of adenine to guanine which results in a C29R amino acid substitution in the DPD enzyme whose normal function catalyzes the rate-limiting step of thymine, uracil, and fluoropyrimidine metabolism, thereby enabling the metabolism of 5-FU [12]. This abnormal function caused by the *DPYD**9A, 85 A>G SNP means that 5-FU cannot be effectively eliminated by the body leading to a level of accumulation within the body that induces neutropenia toxicity [13].

*DPYD**9A, 85 A>G showed a 100% (2/2) rate of toxicity including leukopenia and neutropenia from the first cycle of 5-FU for homozygous variants in this study. This suggests a very high association between patients carrying the homozygous variant (GG) of *DPYD**9A, 85 A>G and neutropenia occurring when receiving their first 5-FU cycle, which should be a cause of concern in the clinical setting. This finding is in accordance with existing research of Ashok V. et al., who reported that colorectal cancer patients with the homozygous variant (GG) of *DPYD**9A, 85 A>G showed significantly higher concentrations of 5-FU from the first cycle of capecitabine administration [14]. Patients receiving *DPYD** 9A heterozygous genotype had an incidence of middle-severe nausea and vomiting higher than *DPYD** 9A wild genotype [15]. Furthermore, *DPYD**9A variant was associated with diarrhea in metastatic gastrointestinal patients treated with capecitabine-based chemotherapy [16]. Cancer patients with *DPYD* 85 A>G (C29R) mutation showed significantly reduced DPD activity [17]. Bain BJ. et al., reported that homozygous and heterozygous of *DPYD**9A had no effect in term of 5-FU plasma clearance, uracil concentration, UH2/U ratio and toxicity, while one patient with variant of *DPYD**9A, IVS14+1G>A, and 2846 A>T was related to grade IV polyvisceral toxicity and died after 40 days [18].

In this study, *DPYD**5, 1627 A>G was found to have a significant impact on the likelihood of neutropenia occurring as an adverse side effect of administering a second cycle of 5-FU, though to a lesser extent than that of *DPYD**9A. Although it requires a second cycle for the effects to be significant, it should be noted that neutropenia occurred most predominantly within the heterozygous variant group of *DPYD**5. There were no statistically significant *DPYD**5,1627 A>G which was the most prevalent variant genotype in the cohort of this study. However, we found that 100% (2/2) of patients carrying the homozygous variant (GG) presented no hematological toxicity in a significant manner ($P=0.02$). Similarly, Lay T. et al. have reported that *DPYD* 1627 A>G may potentially be used as a predictive marker for neutropenia in patient treated with 5-FU [19].He YF et al. reported that there was no significant correlation between three mutations [(*DPYD**9A, 85T>C), (*DPYD**5, 1627 A>G) and 1896T>C], and DPD activity [20]. The amino acid substitutions I543V of *DPYD**5,1627 A>G did not significantly influence DPD enzyme activity [21]. However, a variant of *DPYD**5 was related to a lower DPD activity [22]. Thus, patient receiving AA and GA had DPD activity higher than GG patients.

The *DPYD* 1896T>C was found to be a significant factor in determining 5-FU of grade 1–4 leukopenia in the first cycle ($P=0.002$) in this study, another existing study by Lay T. et al., has stated that, within their study, *DPYD* 1896 and *DPYD**5 1627 A>G were responsible for 29.9% of the neutropenia cases ($P=0.017$) [19]. In addition, *DPYD**2A, 1905+1 G>A was also found to be significant in producing 5-FU toxicities in another study by Adam L. et al., in which *DPYD**2A was found to have statistically significant associations with Grade 3 or higher 5-FU adverse events (more specifically, neutropenia, nausea, and vomiting) [10]; however, no allele frequencies were found within this study's sample of colorectal cancer patients. The lack of allele frequencies in *DPYD**2A within this study was similar to that of *DPYD* 1774 C>T for which no variant allele was found either. Moreover, *DPYD* 1774 C>T has been rarely studied among the Asian population and, therefore, evidence regarding its relationship with 5-FU-induced toxicities remains inconclusive. In females, 5-FU is eliminated at a slower rate, leading to higher toxicity. The DPD expression in tumor tissue was significantly lower in females compared to males. This reduced clearance of 5-FU in females highlights

Table 8 Multivariate analysis of predictive factors for the 5-FU-induced neutropenia to colorectal cancer patients ($N=75$)

Factors	First cycle						Second cycle					
	Grade 1–4 neutropenia[a]			Grade 3–4 neutropenia[b]			Grade 1–4 neutropenia[a]			Grade 3–4 neutropenia[b]		
	Exp (B)	95%CI	*P* value	Exp (B)	95%CI	*P* value	Exp (B)	95%CI	*P* value	Exp (B)	95%CI	*P* value
Age (years)	1.00	0.96–1.06	0.819	1.01	0.94–1.08	0.845	1.00	0.96–1.05	0.956	0.96	0.90–1.01	0.154
Gender	1.90	0.59–6.13	0.284	4.62	0.86–24.93	0.075	1.65	0.57–4.79	0.357	1.44	0.32–6.48	0.634
ECOG performance status	1.50	0.53–4.22	0.442	1.42	0.30–6.70	0.654	1.52	0.60–3.80	0.373	0.66	0.15–2.84	0.578
Sites of metastases	0.808	0.53–1.24	0.332	0.86	0.48–1.54	0.621	0.84	0.57–1.23	0.367	1.28	0.75–2.19	0.364
Line of treatment	0.230	0.54–0.98	0.230	0.19	0.02–2.32	0.192	0.73	0.27–1.97	0.536	0.35	0.04–2.87	0.328
*DPYD*5 (c.1627 A > G)*	0.436	0.15–1.27	0.127	0.34	0.06–1.90	0.220	1.08	0.45–2.62	0.862	1.49	0.43–5.18	0.531
*DPYD*9A (c.85 A > G)*	3.397	0.84–13.73	0.086	2.33	0.57–9.54	0.239	0.82	0.26–2.56	0.731	0.32	0.03–3.08	0.327

the need for sex-specific dosing of fluoropyrimidines [23]. The increased toxicity in females is due to slower clearance of 5-FU. These findings indicate that fluoropyrimidine dosing should be adjusted according to sex [24]. Similarly, this study reported that females had significantly higher neutropenia than males. However, multivariate analysis revealed no significant association between clinical characteristics and genetic polymorphisms and neutropenia in Thai colorectal cancer treated with 5-FU.

The limitations of this study include its small sample size as well as the fact that non-hematological toxicities (e.g., severe diarrhea) were not assessed as part of the side effects of 5-FU. As such, a replication study cohort would be needed for further study. Moreover, our study was limited to *DPYD*, whereas other genetic polymorphisms have been shown to produce significant 5-FU toxicities in colorectal patients, such as those in the thymidylate synthetase gene (*TYMS*) [25] and the Methylenetetrahydrofolate reductase gene (*MTHFR*) [26]. Further study, dose reduction would be analysis for patients who experience toxicity.

In conclusion, two out of the five SNPs explored in our study were found to be associated with adverse side effects after the administration of 5-FU in Thai colorectal cancer patients, namely, *DPYD**9A, 85 A > G and *DPYD**5, 1627 A > G. The adverse side effects that were assessed in this study were related to hematological toxicities of which only neutropenia was found to present a significant relationship with genetic polymorphisms of the *DPYD* gene. Our findings suggest the existence of an important link between *DPYD* genetic polymorphisms and the side effects of 5-FU in Thai colorectal cancer patients, making genotyping for *DPYD* variants prior to 5-FU therapy an appropriate recommendation.

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Declarations

Competing interests The authors declare no competing interests.

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